

Project Report: Automated Network Request Management in ServiceNow

Project Title: Automated Network Request Management in ServiceNow

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GitHub Repository: <https://github.com/ZEESHAN55555/service-now/upload/main>



Automated Network Request
Management In Servicenow

Abstract

This project demonstrates the automation of **network request management** within ServiceNow using **Service Catalogs, Variable Sets, Flow Designer, Approvals, and Notifications**.

The system allows users to raise a network-related request through a **catalog item**, capture all necessary inputs via **variables**, and process the request automatically using **flow automation**. The project eliminates manual intervention, reduces approval delays, and ensures real-time communication with stakeholders through email notifications.

Introduction

ServiceNow is a powerful cloud-based ITSM (IT Service Management) platform. One of its most impactful features is the ability to **automate workflows**, reducing repetitive tasks and ensuring standardization.

In IT organizations, **network requests** (VPN access, firewall changes, port requests, etc.) are frequent. Manual handling of such requests is prone to delays, miscommunication, and lack of tracking.

This project solves that by:

1. Creating a **Catalog Item** for submitting network requests.
2. Designing a **Flow** in Flow Designer to automatically process requests.
3. Implementing **approvals** for supervisor authorization.
4. Sending **automated email notifications** to requesters.

Objectives

1. **Automation** – Replace manual request processing with fully automated workflows.
2. **Standardization** – Ensure every request follows a structured and auditable path.
3. **Efficiency** – Reduce delays caused by manual approvals and notifications.
4. **User Experience** – Provide a simple and professional request submission form.
5. **Transparency** – Keep all stakeholders informed of request status in real time.

Literature Review

1. **Service Catalog** – A digital catalog containing services available for request. Each item can have fields (variables) to capture user input.
2. **Variable Set** – A reusable group of variables that can be applied to multiple catalog items.
3. **Flow Designer** – A no-code automation tool in ServiceNow for creating process flows.
4. **Approvals** – Used to enforce authorization before fulfilling requests.
5. **Notifications** – Automated emails or messages sent at key stages of the workflow.

This project combines all of these ServiceNow concepts into a single automated solution.

Methodology

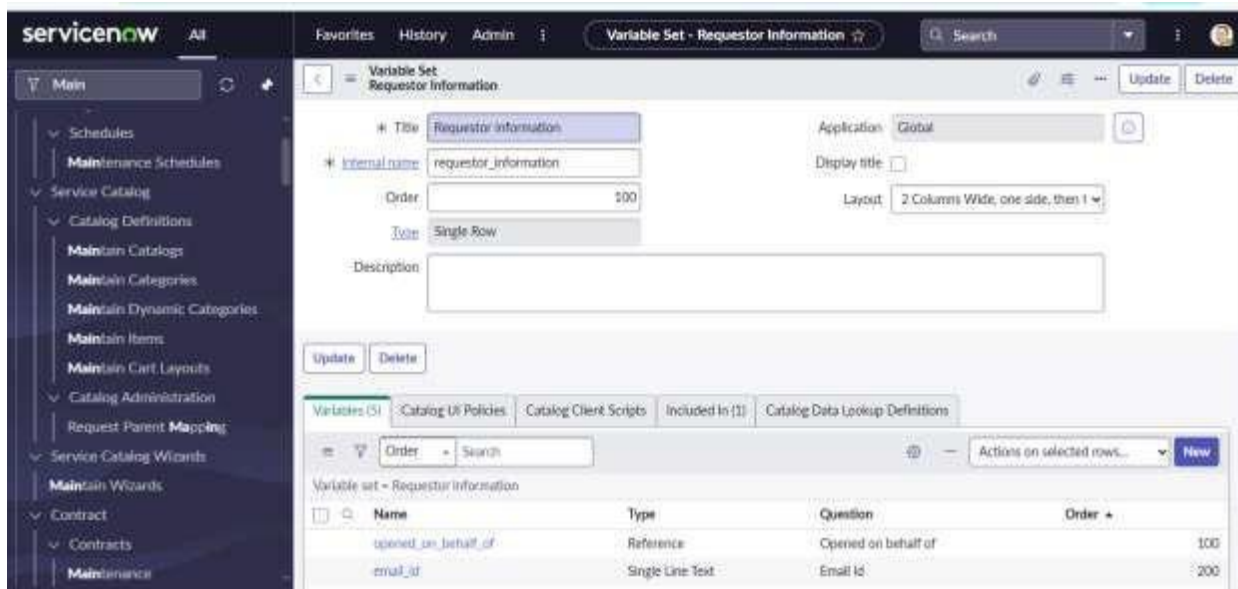
The project follows a **step-by-step approach**:

Step 1: Service Catalog Item Creation

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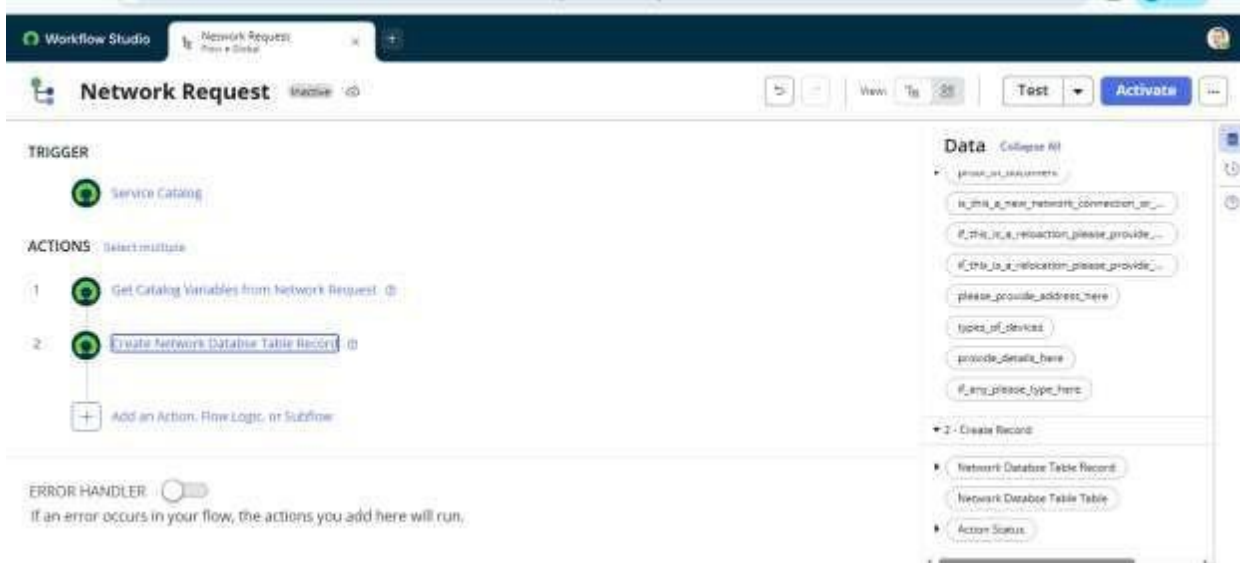
A **Variable Set** was configured to capture inputs like:

1. Request Type (VPN, Firewall, Port Access, etc.)
2. Requester Details (Name, Department, Email)
3. Justification



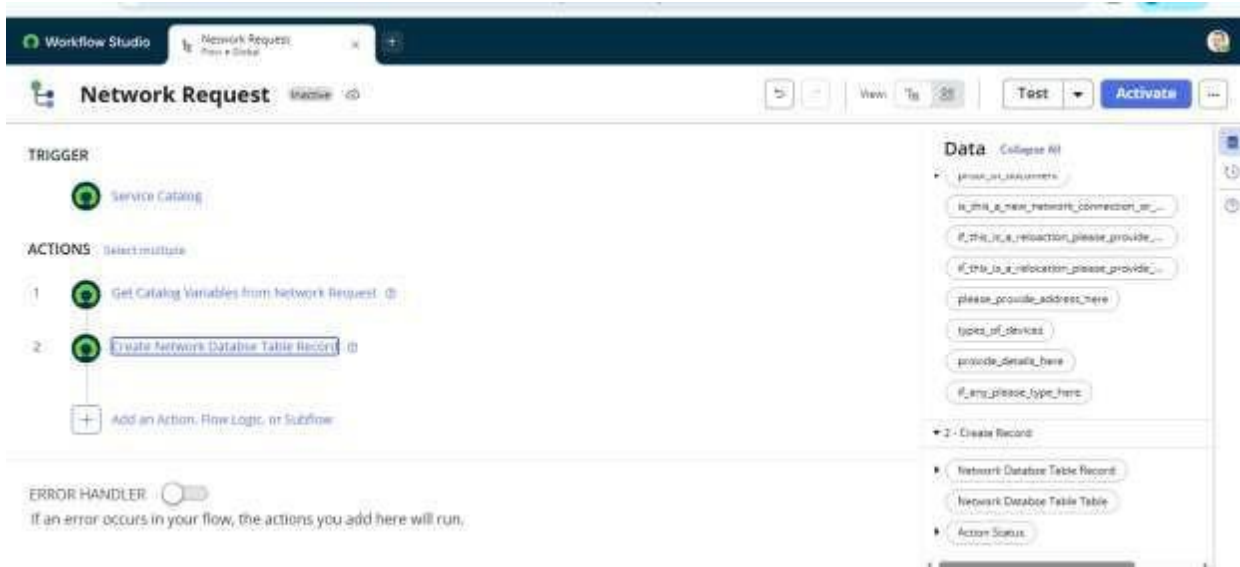
Step 3: Flow Designer – Initial Setup

A new flow was created in Flow Designer triggered when the catalog item is submitted.



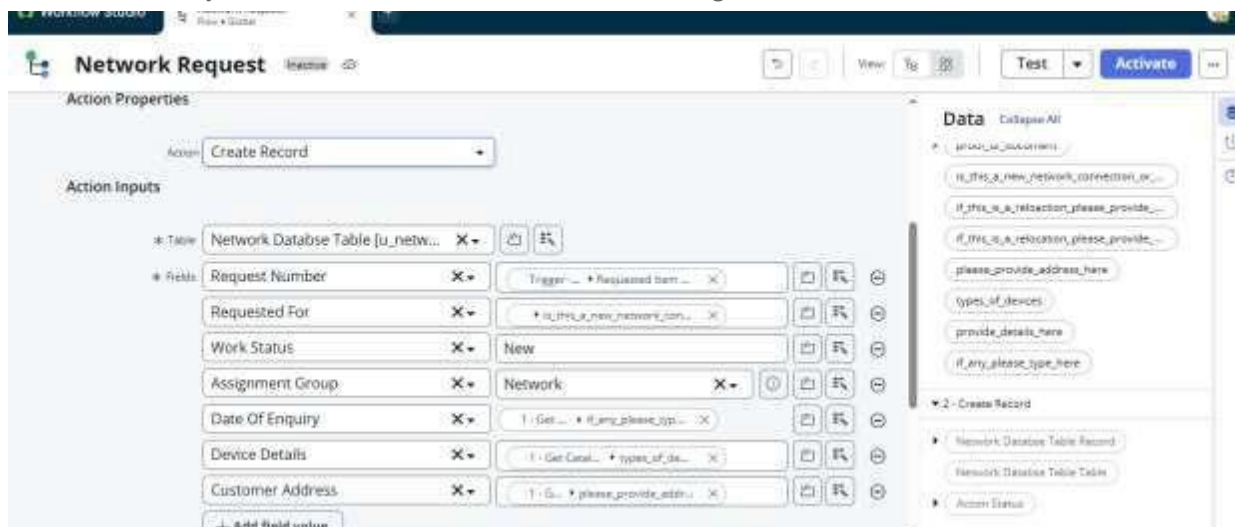
Step 4: Get Catalog Variables Action

This step fetches the submitted variables for further processing.



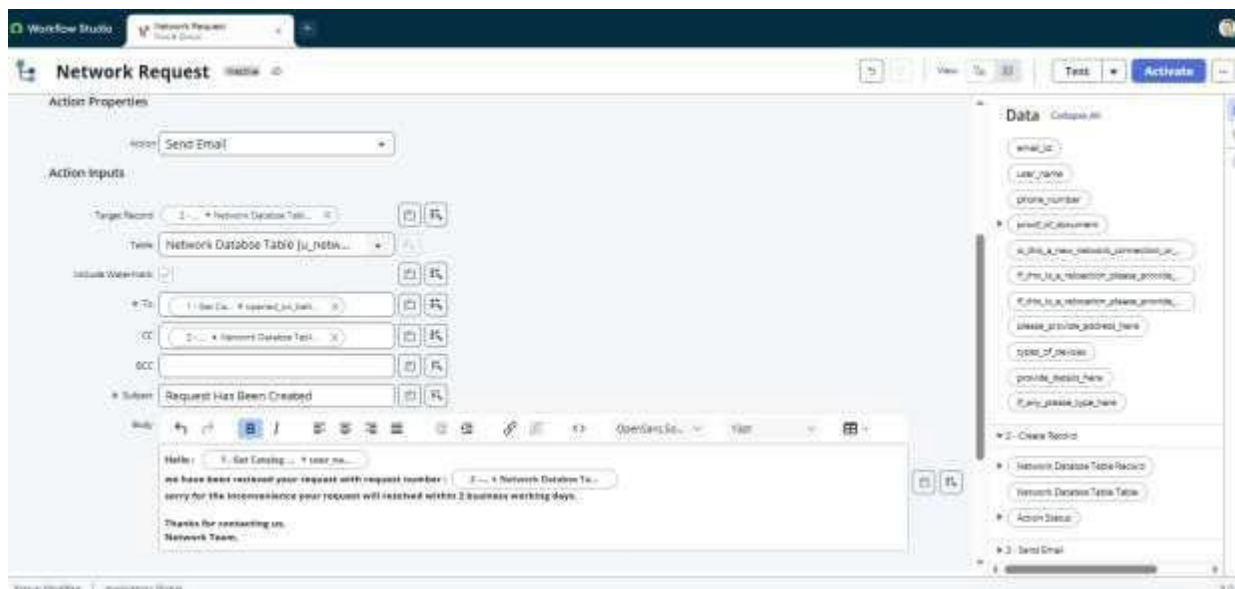
Step 5: Create Record Action

Automatically creates a record in the database for tracking.



Step 6: Send Email Action

An automated email notification is sent to the requester with request details.



Step 7: Ask for Approval Action

The request is routed to the **Approver (Manager/Network Admin)** for authorization.

The screenshot displays the 'Network Request' form in the ServiceNow interface. The form is divided into several sections:

- Action:** A dropdown menu set to 'Ask For Approval'.
- Condition Inputs:**
 - Record:** A dropdown menu set to 'Network Database Table'.
 - Table:** A dropdown menu set to 'Network Database Table (u_netw...'.
 - Approval Reason:** A dropdown menu set to 'Waiting For Approval'.
 - Approval Field:** A dropdown menu set to 'Select a field'.
 - Journal Field:** A dropdown menu set to 'Select a field'.
- Rules:**
 - Approve:** A dropdown menu set to 'Approve'.
 - Anyone approves:** A dropdown menu set to 'Anyone approves'.
 - When:** A dropdown menu set to '2. Cre... • ... • Assignment G...'.
 - Logic:** Buttons for 'OR' and 'AND' logic.
- Due Date:** A dropdown menu set to 'None'.

On the right side, there is a **Data** section with a list of fields: email_id, user_name, phone_number, proof_of_document, is_this_a_new_network_connection_or..., if_this_is_a_relocation_please_provide..., if_this_is_a_relocation_please_provide..., please_provide_address_here, types_of_devices, provide_details_here, and if_any_please_type_here. Below the list is a button labeled '2 - Create Record'.

Step 8: Conditional Flow Logic

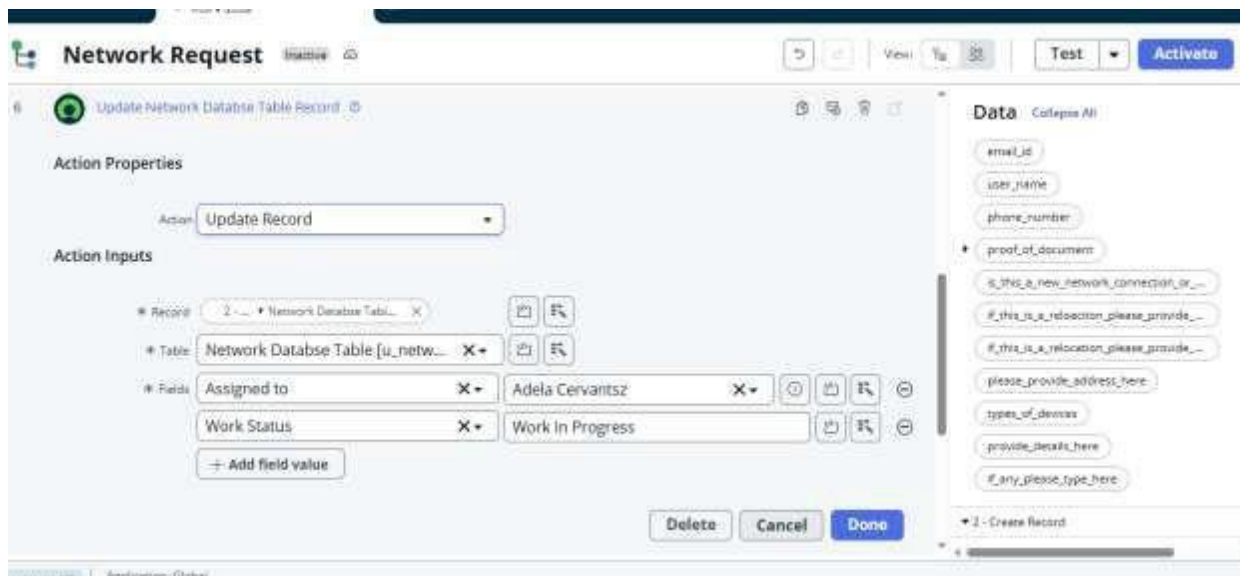
Based on approval:

1. **If Approved** → Proceed with request fulfillment.
2. **If Rejected** → Notify requester with reason.

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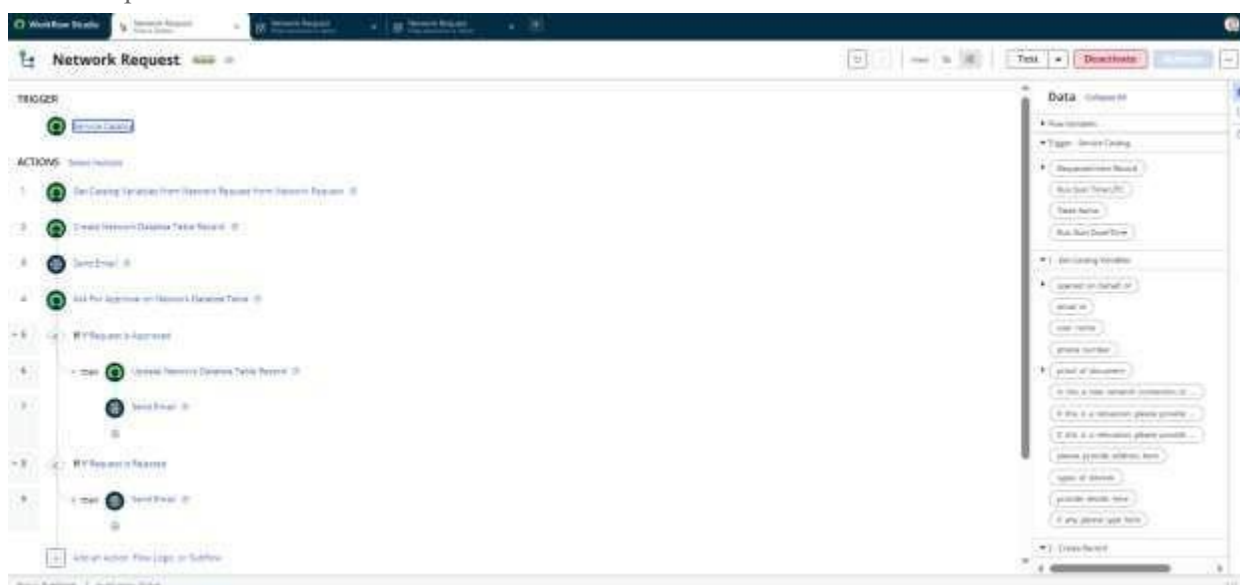
Step 9: Update Record Action

The record is updated with the latest status (Approved/Rejected).



Step 10: Final Flow Design

The complete workflow is now automated end-to-end.



Results

1. Users can now raise network requests directly from the catalog.
2. Each request is automatically logged, tracked, and routed.
3. Approvals and notifications are **fully automated**.

4. Manual effort and errors are significantly reduced.

The screenshot shows the ServiceNow interface for a 'Network Request'. The breadcrumb trail is: Home > Service Catalog > Standard Changes > Network Standard Changes > Network Request. The form includes fields for 'Opened on behalf of' (Abraham Lincoln), 'Phone Number' (0966004403), 'Email id' (abraham.lincoln@example.com), and 'User name' (abraham). There is an 'Upload' button for 'Proof of Document'. On the right, there are controls for 'Quantity' (1), 'Delivery Time' (0 Days), and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. At the bottom left, there are radio buttons for 'New', 'Relocation', and 'None' under the heading 'Is this a new network connection or a relocation?'. A chatbot icon is visible in the bottom right corner.

Conclusion

The project successfully demonstrates the **automation of network requests in ServiceNow**. By leveraging **Catalogs, Variables, Flow Designer, Approvals, and Notifications**, the system ensures faster resolution, transparency, and standardization.

This implementation can be scaled across other IT service requests like hardware provisioning, software installations, and employee onboarding.

Future Enhancements

1. **Integration with CMDB** – Automatically update configuration items linked to requests.
2. **SLA Tracking** – Attach Service Level Agreements for timely fulfillment.
3. **Chatbot Integration** – Allow employees to raise requests via a virtual agent.
4. **Reporting Dashboard** – Provide real-time analytics on request trends.

References

1. ServiceNow Documentation – <https://docs.servicenow.com/>
2. GitHub Project Repository – <https://github.com/ZEESHAN55555/service-now/upload/main>